

Reproducibility of the Flagship Sweep

1 Summary

We evaluated whether the completed sweep in `runs/flagship/` reproduces Figure 2 of *Slow Transition to Low-Dimensional Chaos in Heavy-Tailed Recurrent Neural Networks*. The answer depends on which Lyapunov analysis is used:

1. **The published numerical transition points are closely reproduced when we emulate the original study’s historical analysis errors.** This analysis propagated tangent vectors with a transposed Jacobian, $J^\top Q$, and evaluated the Jacobian one state transition too late. It reproduces 7 of the paper’s 9 transition grid points exactly; the other 2 differ by one adjacent gain value.
2. **The results change when the Lyapunov analysis is corrected.** The canonical update, JQ , with the Jacobian evaluated at the state that generates the transition produces substantially earlier transitions in the most heavy-tailed conditions. It exactly matches only 5 of the 9 reported grid points and accumulates 11 grid steps of absolute transition error, compared with 2 for the historical analysis.

Thus, the paper’s qualitative conclusion—a slower, broader transition to chaos for heavier-tailed networks—remains visible under the corrected analysis. The paper’s reported quantitative transition locations, however, appear to reflect the historical, erroneous Lyapunov calculation.

2 Target experiment

The comparison targets Figure 2 of the archived paper, which reports the maximum Lyapunov exponent as a function of recurrent gain for autonomous networks with:

- $N \in \{1000, 3000, 10000\}$;
- $\alpha \in \{1.0, 1.5, 2.0\}$;
- 50 logarithmically spaced gains from 0.01 to 10;
- 10 random trials per condition;
- 2,900 warmup steps followed by 100 Lyapunov accumulation steps; and
- the leading 100 Lyapunov exponents.

These settings are reproduced by `configs/flagship.yaml`. The completed run contains all 4,500 expected condition/trial artifacts. Every recorded metric is finite, no QR stretch floors were used, and the 52 non-CUDA tests pass.

The transition estimator follows the paper’s plotting code: for each (N, α) , trial MLEs are averaged at every sampled gain, and g^* is the first gain where the mean MLE changes from negative to nonnegative. Because the gain grid is logarithmic, adjacent values differ by a factor of approximately 1.151. A one-grid-step disagreement therefore corresponds to about a 15% difference in the reported discrete threshold.

3 Correct and historical Lyapunov analyses

For the autonomous recurrence

$$h_{t+1} = \tanh(Wh_t),$$

the Jacobian for the transition from h_t to h_{t+1} is

$$J_t = D_t W, \quad D_t = \text{diag}(\text{sech}^2(Wh_t)).$$

The canonical forward tangent update is

$$Q_{t+1} R_{t+1} = J_t Q_t.$$

This is the algorithm stated in the paper and implemented by the corrected code in `simulation.py`.

The state update and its Jacobian use the same preactivation.

The archived study implementation historically made two changes to this calculation:

1. **Transposed tangent propagation.** A comment in `original/LypAlgo.py` records that the code used $J^\top Q$ before July 8, 2026.
2. **One-step-late Jacobian indexing.** The loop first advances `ht` to h_{t+1} and then passes that updated state to `rnn_jac` (update, Jacobian call). `rnn_jac` subsequently recomputes Wh_{t+1} (construction), so the tangent update uses J_{t+1} rather than J_t .

The historical update was therefore effectively

$$Q_{t+1} R_{t+1} = J_{t+1}^\top Q_t.$$

This is not equivalent to the canonical calculation for a time-varying Jacobian sequence. In particular,

$$J_T^\top \cdots J_1^\top = (J_1 \cdots J_T)^\top,$$

which does not preserve the chronological product $J_T \cdots J_1$. The difference is especially consequential for finite-time Lyapunov estimates in nonlinear, strongly non-normal heavy-tailed networks.

4 Finding 1: the historical analysis closely reproduces Figure 2

We regenerated the weights from the saved condition seeds and re-evaluated the saved trajectories using the historical calculation. The re-analysis used all 100 tangent directions, 100 float32 QR steps, the original 10^{-12} QR floor, the maximum of the resulting 100 exponents per trial, and the mean over 10 trials. This matches the analysis dimensions used for Figure 2 while holding the networks and trajectories fixed across analysis variants.

N	α	Paper g^*	Historical $J_{t+1}^\top Q$	Correct $J_t Q$
1,000	1.0	0.791	0.910	0.391
1,000	1.5	0.791	0.687	0.596
1,000	2.0	0.910	0.910	0.910
3,000	1.0	0.450	0.450	0.295
3,000	1.5	0.596	0.596	0.596
3,000	2.0	0.910	0.910	0.910
10,000	1.0	0.295	0.295	0.256
10,000	1.5	0.518	0.518	0.518
10,000	2.0	0.791	0.791	0.791

The corresponding transition metrics are:

Analysis	Exact paper grid matches	Total absolute grid-step error
Correct $J_t Q$	5/9	11
$J_t^\top Q$, correct indexing	6/9	3
Historical $J_{t+1}^\top Q$	7/9	2

The two remaining historical-analysis differences are both adjacent grid points at $N = 1000$. They are much smaller than the discrepancies produced by the corrected analysis. At $N = 1000, \alpha = 1$, for example, the historical mean MLE is -0.0062 at the paper's $g = 0.791$ and $+0.0118$ at the next gain, $g = 0.910$. The reported grid point is therefore sensitive to the exact set of ten random realizations.

5 Finding 2: the corrected analysis deviates from Figure 2

At four informative paper boundaries, changing from the corrected analysis to the correctly indexed transposed analysis changes the mean MLE as follows:

N	α	g	Correct $J_t Q$	$J_t^\top Q$
1,000	1.0	0.791	+0.1470	-0.0002
1,000	1.5	0.791	+0.0958	+0.0167
3,000	1.0	0.450	+0.0985	+0.0131
10,000	1.0	0.295	+0.0496	-0.0008

The transpose changes the mean MLE by roughly 0.05 to 0.15 in these heavy-tailed conditions. By comparison, shifting the Jacobian index from the correct J_t to the historical J_{t+1} changes the same means by only about 0.002 to 0.006. The indexing error is nevertheless sufficient to move $N = 10000, \alpha = 1$ from $g^* = 0.339$ to the paper's $g^* = 0.295$, because the mean lies extremely close to zero.

6 Why the largest discrepancies occur for small heavy-tailed networks

The corrected and historical analyses disagree most for $N = 1000$ with $\alpha = 1.0$ or 1.5 . Three effects reinforce one another there.

6.1 Nonlinear and heterogeneous Jacobian gating

Writing $J_t = D_t W$ makes the orientation error explicit:

$$J_t Q = D_t W Q, \quad J_t^\top Q = W^\top D_t Q.$$

The two updates gate opposite sides of the recurrent matrix. At the paper’s reported gains, the $N = 1000$ heavy-tailed networks contain substantially more saturated activity, so D_t is heterogeneous and far from the identity:

N	α	Mean $1 - h^2$	Fraction with $ h \geq 0.99$
1,000	1.0	0.738	7.43%
3,000	1.0	0.932	1.66%
10,000	1.0	0.987	0.29%
1,000	1.5	0.810	1.72%
3,000	1.5	0.955	0.31%
10,000	1.5	0.984	0.12%

At larger N , the heavy-tail transition occurs at a lower gain while the activity remains closer to the quiescent state. Consequently $D_t \approx I$, and the orientation error has less effect on the detected crossing. Gaussian networks also lack the rare, dominant connections that make heavy-tailed Jacobians strongly heterogeneous and non-normal.

6.2 Weak self-averaging

For $\alpha < 2$, rare extreme weights dominate individual network realizations. Transition locations therefore concentrate slowly with network size. In the corrected sweep, the standard deviation of the per-trial crossing index was 1.64, 1.55, and 0.46 grid steps for $\alpha = 1$ at $N = 1000, 3000, 10000$, respectively. For $\alpha = 1.5$, the corresponding values were 0.87, 0.75, and 0.50. Gaussian values were only 0.40–0.49 grid steps.

6.3 Coarse thresholding and different random streams

The paper reports the first nonnegative point on a 50-point gain grid rather than an interpolated root. Small changes around zero therefore move the reported result by a full 15% grid step. The original study used a sequential SciPy/global RNG stream beginning with `seed + trial` (`fig2_parallel.py`); this reimplement uses independently keyed condition streams. Both sample the same target distributions, but they do not generate the same ten matrices.

This distinction matters most for the two $N = 1000$ heavy-tail cases. Under the historical re-analysis, a bootstrap over the current ten $N = 1000, \alpha = 1$ trial curves selected the paper’s $g = 0.791$ crossing about 20% of the time. The archived submission script also specifies only $N = 1000$ and trial 0 (`submit_fig2_parallel.sh`), not the three sizes and ten trials reported in the paper, so it does not retain the exact execution provenance of Figure 2.

7 Interpretation

The completed flagship sweep supports two distinct reproducibility claims:

- **Historical numerical reproduction:** yes. When the historical transpose and indexing behavior are restored, the reported Figure 2 transition points are recovered to within one sampled gain everywhere, with exact agreement in 7 of 9 conditions.
- **Reproduction under the algorithm stated in the paper:** no, not quantitatively. The corrected $J_t Q$ analysis yields materially earlier transitions for heavy-tailed networks, most prominently at $N = 1000, \alpha = 1$, $N = 1000, \alpha = 1.5$, and $N = 3000, \alpha = 1$.

The corrected sweep still reproduces the qualitative scientific pattern: heavier tails yield a slower, broader approach to chaos, and increasing network size shifts the transition toward lower gain. What changes is the quantitative location and magnitude of that transition. Accordingly, results from runs/flagship/ should be described as a **corrected-analysis replication**, while the historical re-analysis strongly suggests how the paper's published numerical values arose.